

Resource Allocation System

Iteration 28 – Multi-Room Exam Greedy Strategy

Najib Attar
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1 Iteration Goal

The goal of Iteration 28 is to introduce the first real room-pool sharing strategy for exam allocation. Earlier iterations prepared the model by adding `ExamRequest`, adding `assignedParticipants` to `Allocation`, and allowing allocation strategies to receive the full list of loaded spaces.

This iteration adds `MultiRoomExamGreedyStrategy`. The strategy allows a splittable exam request to be assigned across multiple classrooms when one classroom is not enough. This does not mean placing two exams in the same room at the same time. Instead, it means that one exam may use several available classrooms from the same room pool.

2 Changes from Iteration 27

Iteration 27 prepared the allocation model for future multi-room work. It separated total request demand from room-level assignment by adding `assignedParticipants` to `Allocation`. It also updated the allocation strategy interface so strategies receive the full `spaces` list.

Iteration 28 uses those preparations by adding a concrete strategy that can allocate a splittable `ExamRequest` across more than one classroom. The existing greedy and priority strategies remain available and unchanged in behavior.

The main changes are:

- Added `MultiRoomExamGreedyStrategy.h`.
- Added `MultiRoomExamGreedyStrategy.cpp`.
- Updated `AllocationStrategyFactory` to support `multi_room_exam_greedy`.
- Updated config validation so `data/config.txt` can select the new strategy.
- Updated `data/requests.csv` with sample splittable exam requests.
- Preserved normal greedy behavior for non-exam requests.

3 Scenario

A large exam may have more students than any single classroom can hold. For example, a final exam may require seats for 70 students, while the available classrooms have capacities of 40 and 35.

With the new strategy, the system can create multiple allocations for the same exam request:

Request	Room	Capacity	Assigned Participants
24	B201	40	40
24	B202	35	30

The total assigned participants is 70, so the exam can be approved. If the available classroom pool cannot satisfy the full participant count, the request is rejected and no partial allocations are committed.

4 Modules and Their Tasks

Module	Task in Iteration 28
<code>MultiRoomExamGreedyStrategy</code>	Implements the new multi-room greedy allocation behavior for splittable exam requests. It also delegates normal non-exam processing to the existing greedy behavior.
<code>IAllocationStrategy</code>	Already receives requests, spaces, allocations, and the rule engine facade. The new strategy implements this existing interface.
<code>AllocationStrategyFactory</code>	Returns <code>MultiRoomExamGreedyStrategy</code> when the selected strategy name is <code>multi_room_exam_greedy</code> .
<code>DataController</code>	Accepts <code>multi_room_exam_greedy</code> as a valid strategy value from the config file.
<code>ExamRequest</code>	Provides exam-specific data, including <code>canSplitAcrossRooms</code> and the exam time slot.
<code>Allocation</code>	Stores the room-level <code>assignedParticipants</code> value for each created allocation.
<code>AllocationWriter</code>	Continues exporting allocations, including the assigned participant count for each room.

5 Multi-Room Exam Greedy Design

The new strategy processes requests in loaded order, similar to the normal greedy strategy. For non-exam requests, it uses the existing greedy processing behavior. For exam requests

that cannot split across rooms, it also behaves like the existing one-room exam allocation. For splittable **ExamRequest** objects, the strategy uses the full spaces pool. Candidate rooms must satisfy these conditions:

- The space type must be **Classroom**.
- The room must be available and not under maintenance.
- The room must match the required building when a building is required.
- The room must contain the required feature when a feature is required.
- The requester must be allowed to request the classroom space type.
- The room must not already have an overlapping allocation at the exam time.

Rooms are selected greedily in the order loaded from `spaces.csv`. No best-fit sorting is applied in this iteration.

The strategy builds a temporary assignment list first. Each room receives:

$$\text{assignedParticipants} = \min(\text{room capacity}, \text{remaining participants})$$

Only after the total assigned participants reaches the exam participant count does the strategy approve the request and commit all allocations. If the total available capacity is not enough, the request is rejected and no partial allocations are created.

6 Iteration Comparison

Iter.	Focus	Added / Modified	Design Responsibility	Responsibility	Prototype Effect
24	Request meta-data	title and purpose in Request	Added descriptive request information at the base request level.		Requests became clearer and better prepared for exam-related extensions.
25	Exam request model	ExamRequest	Represented exams as a first-class request type instead of only a purpose value.		Exam-specific fields could be loaded, printed, and exported.
26	Request-type permission	RequestTypeRule	Separated request-type permission from space-type authorization.		Students and staff could be rejected from submitting exam requests.
27	Allocation preparation	assignedParticipants and strategy space pool input	Prepared allocations and strategies for future multi-room exam work.		Strategies received enough information to choose from all loaded spaces.
28	Multi-room exam greedy	MultiRoomExamGreedyStrategy	Substituted the first room-pool sharing allocation strategy for split-table exams.		A large exam can now be approved across multiple classrooms when total capacity is enough.

7 Updated Class Diagram

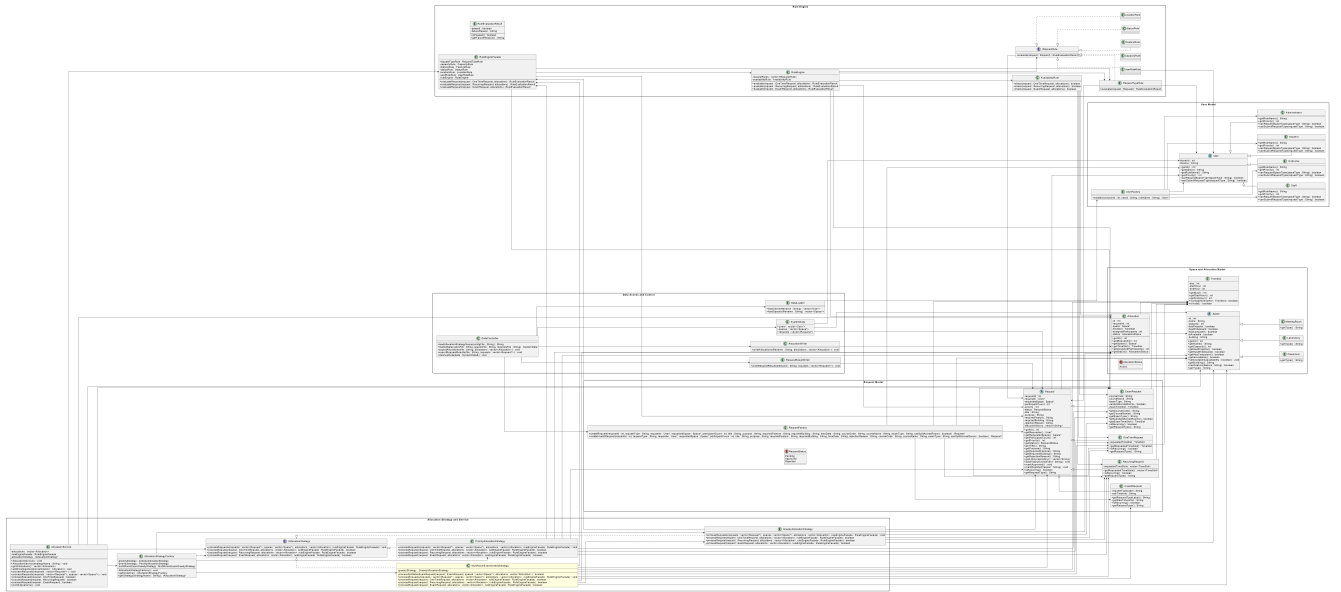


Figure 1: Updated class diagram after adding MultiRoomExamGreedyStrategy.

8 Updated Workflow Diagram

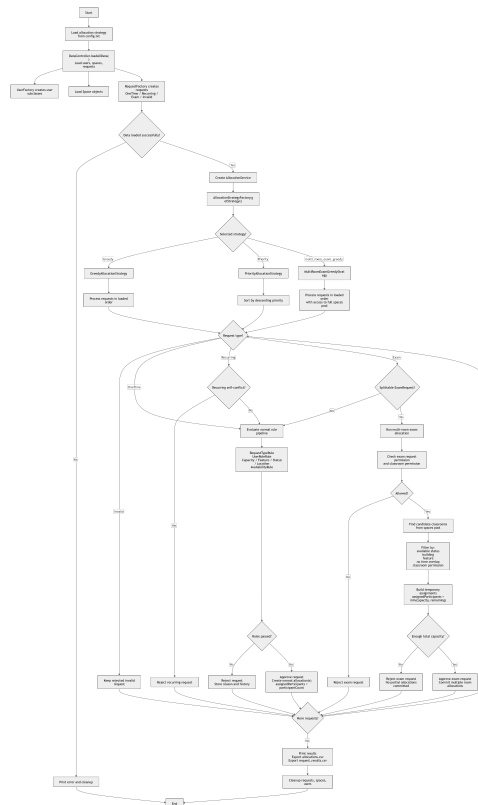


Figure 2: Updated workflow for multi-room exam greedy allocation.

9 Prototype Implementation

The new strategy can be selected externally through the config file:

```
allocation_strategy=multi_room_exam_greedy
```

When this strategy is selected, the allocation service delegates batch processing to `MultiRoomExamGreedy`. The strategy handles normal requests through existing greedy behavior, but applies multi-room allocation logic to splittable exam requests.

The prototype data includes two important examples:

- Request 24 is a splittable exam with 70 participants. It is approved using two classrooms: B201 receives 40 participants and B202 receives 30 participants.
- Request 25 is a splittable exam with 120 participants. It is rejected because the available classroom capacity is not enough, and no partial allocation rows are created.

The generated `allocations.csv` output shows multiple allocation rows for the approved split exam. Each row records the room and its assigned participant count.

10 Summary

Iteration 28 adds the first concrete multi-room exam allocation strategy. The system can now use the classroom pool to satisfy a splittable exam request across multiple rooms while preventing overlapping use of the same room.

This iteration does not implement best-fit selection, priority-based multi-room allocation, same-room sharing, or partial room sharing between different exams. It keeps the algorithm simple and focused: choose valid classrooms greedily, confirm enough total capacity, then commit all allocations together.

The result is an important step toward full exam allocation support while preserving the existing rule engine, request factory, user permissions, allocation exports, and previous greedy and priority strategies.